

IT2011 - Artificial Intelligence and Machine Learning

Final Evaluation - Implementation

Individual Model Report

Letter Recognition using Support Vector Machine (SVM)

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Group	_____
Model	Support Vector Machine (SVM)
Problem Type	Multiclass Classification
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Abstract

This report presents the individual Support Vector Machine (SVM) implementation for an English letter recognition project. The task is a multiclass classification problem with 26 target classes representing letters A to Z. The group preprocessing pipeline provided standardized numerical features and a PCA version of the same data. Three SVM kernels were tested first: Linear, Radial Basis Function (RBF), and Polynomial. RBF was then compared on scaled features and PCA features. A small GridSearchCV was used to tune the RBF model using C values of 1 and 10 and gamma values of scale and 0.1 with 3-fold cross-validation. The best tuned model used an RBF kernel with C=10 and gamma=0.1. It achieved 96.56% mean cross-validation accuracy and 97.24% final test accuracy. These results show that the tuned RBF SVM was the strongest SVM configuration tested for this dataset.

1. Introduction

Letter recognition is a classification problem where a machine learning model predicts which alphabetic character is represented by a set of numerical measurements. In this project, the target contains 26 classes corresponding to the English letters A to Z. My individual contribution is the design, implementation, tuning, and evaluation of a Support Vector Machine classifier using the datasets produced by the group preprocessing pipeline.

The purpose of this part is not to repeat preprocessing. Instead, the prepared train and test datasets are loaded directly and used to compare different SVM configurations. The implementation is intentionally kept simple so that each step can be demonstrated and explained clearly.

2. Objectives

- Train an SVM model for 26-class English letter classification.
- Compare Linear, RBF, and Polynomial SVM kernels.
- Compare RBF SVM performance using standardized features and PCA features.
- Tune the RBF SVM using a small GridSearchCV and 3-fold cross-validation.
- Evaluate the best model using accuracy, precision, recall, F1-score, and a confusion matrix.
- Save the final trained SVM model for later use.

3. Dataset and Prepared Inputs

The preprocessing stage was completed before this individual SVM implementation. Therefore, the SVM code uses already prepared CSV files. This avoids repeating the train/test split or scaling inside the model notebook.

Dataset	Rows	Input Dimensions	Purpose
Scaled Train	14,934	16 features	SVM training and tuning
Scaled Test	3,734	16 features	Final SVM evaluation
PCA Train	14,934	12 components	PCA comparison
PCA Test	3,734	12 components	PCA comparison

The target column is named letter. The class labels are stored as integers from 0 to 25, which correspond to A to Z when the classification report is displayed. The scaled dataset contains 16 standardized numerical input features. The PCA dataset contains 12 principal components.

4. Why Support Vector Machine is Suitable

Support Vector Machine is appropriate for this task because it is a supervised classification algorithm that works effectively with numerical features. SVM attempts to create decision boundaries that maximize the margin between classes. Although the original SVM formulation is binary, scikit-learn handles multiclass classification internally for SVC.

- The input data is numerical and already standardized.
- The problem is supervised multiclass classification.
- The dataset size is practical for SVC training.
- Different kernels allow both linear and nonlinear decision boundaries.
- The RBF kernel can model more complex relationships between letter features.

5. Libraries and Main Functions

Library / Function	Use in Project	Reason
pandas	Read CSV files and create result tables	Simple data handling
matplotlib	Display the confusion matrix and comparison chart	Visual evaluation
sklearn.svm.SVC	Create Linear, RBF, and Polynomial SVM classifiers	Main classification algorithm
GridSearchCV	Test small combinations of C and gamma	Hyperparameter tuning
accuracy_score	Calculate overall correct prediction rate	Main comparison metric
classification_report	Show precision, recall, and F1-score	Detailed class evaluation
confusion_matrix	Count correct and incorrect class predictions	Find confused letters
joblib	Save the best trained model	Reuse the final model

6. Methodology and Implementation

1. Load the scaled and PCA train/test CSV files.
2. Separate the feature columns into X and the target column into y.
3. Train Linear, RBF, and Polynomial SVM models using the scaled training data.
4. Compare the RBF model on scaled features with the same RBF model on PCA features.
5. Use GridSearchCV on the scaled training data to tune the RBF model.
6. Select the best tuned estimator and evaluate it on the independent test set.
7. Display class-level metrics, a confusion matrix, and an accuracy comparison chart.
8. Save the final tuned SVM model.

6.1 Feature and Target Separation

```
X_train = scaled_train.drop(columns=["letter"])
y_train = scaled_train["letter"]
X_test = scaled_test.drop(columns=["letter"])
y_test = scaled_test["letter"]
```

X contains the input measurements used by the classifier, while y contains the correct letter class. The test set is kept separate from model fitting and parameter tuning.

6.2 Initial Kernel Comparison

```
linear_model = SVC(kernel="linear", C=1)
rbf_model = SVC(kernel="rbf", C=1, gamma="scale")
poly_model = SVC(kernel="poly", C=1, degree=3, gamma="scale")
```

The first experiment keeps $C=1$ so that the main change is the kernel. This provides a simple comparison of linear and nonlinear SVM decision boundaries.

7. Hyperparameter Tuning

The RBF kernel performed best among the initial scaled-data models, so a small GridSearchCV was used to improve it. Only a small parameter space was used to keep the experiment understandable and computationally manageable.

```
param_grid = {
    "kernel": ["rbf"],
    "C": [1, 10],
    "gamma": ["scale", 0.1]
}

grid = GridSearchCV(SVC(), param_grid, cv=3,
                    scoring="accuracy", n_jobs=-1)
```

Parameter	Values Tested	Meaning
kernel	RBF	Nonlinear radial basis function decision boundary
C	1, 10	Penalty / regularization strength
gamma	scale, 0.1	Influence range of individual training samples
cv	3 folds	Three training/validation splits inside the training data

The grid contains 2 values for C and 2 values for gamma, giving 4 parameter combinations. With 3-fold cross-validation, GridSearchCV performs 12 model fits. The best configuration was $C=10$, $\gamma=0.1$, kernel=RBF, with a mean cross-validation accuracy of 96.56%.

8. Evaluation Metrics

Accuracy: Percentage of all test samples classified correctly. It is useful as the main overall comparison measure.

Precision: For a predicted letter class, precision measures how many of those predictions are correct.

Recall: For an actual letter class, recall measures how many samples were correctly identified.

F1-score: Harmonic balance between precision and recall. It is useful for checking class-level quality.

Confusion Matrix: Shows the number of predictions for every actual/predicted class pair and helps identify which letters are confused.

9. Results

The following table contains the verified results produced by the included dataset and SVM implementation.

Model	Preprocessing	Accuracy (%)
Linear SVM	Scaled (16 features)	83.88
RBF SVM	Scaled (16 features)	93.71
Polynomial SVM	Scaled (16 features)	88.30
RBF + PCA	PCA (12 components)	91.22
Tuned RBF SVM	Scaled + GridSearchCV	97.24

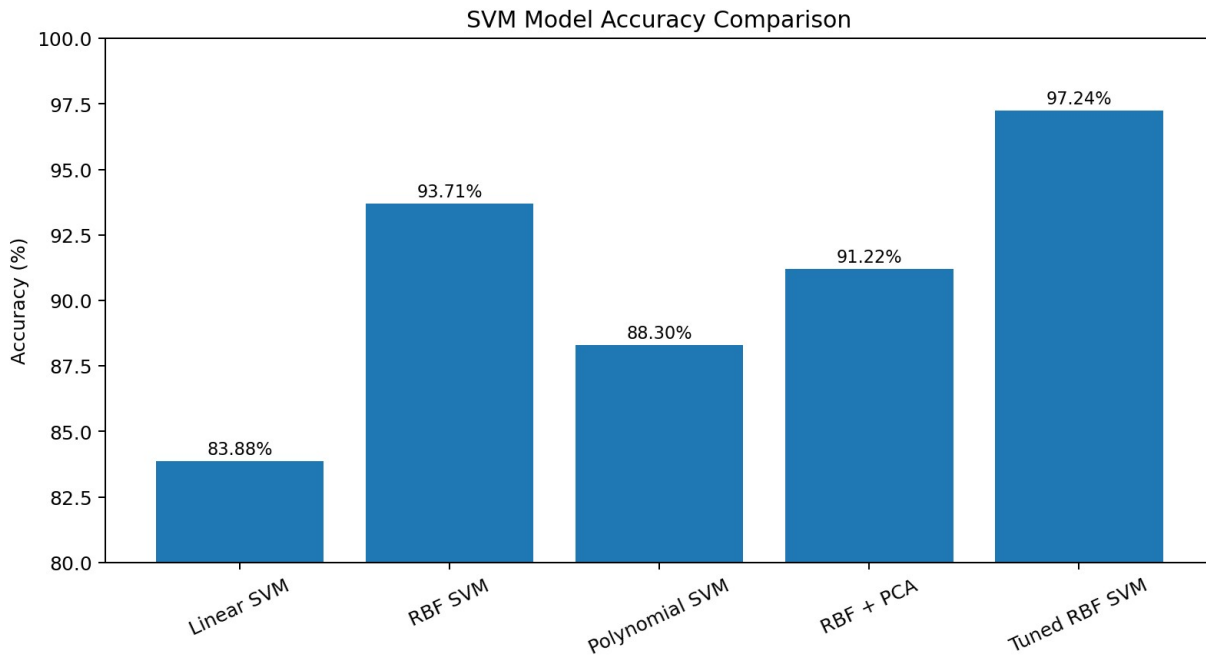


Figure 1. Accuracy comparison of the tested SVM configurations.

The Linear SVM achieved 83.88% accuracy. The RBF SVM improved this substantially to 93.71%, showing that a nonlinear boundary is more suitable for this dataset. The Polynomial SVM reached 88.30%. Using PCA with the RBF model produced 91.22%, which was lower than using all 16 scaled features. After tuning C and gamma, the final RBF SVM reached 97.24% test accuracy, which was the best result of all tested SVM variations.

10. Detailed Performance of the Best Model

The tuned RBF model produced approximately 0.97 macro-average precision, recall, and F1-score across the 26 letter classes. Most classes were recognized with very high accuracy. Class B had the lowest F1-score at about 0.93, followed by H at about 0.94 and R at about 0.95. Several classes achieved precision close to 1.00.

Class	Precision	Recall	F1-score	Support
A	0.99	0.99	0.99	151
B	0.90	0.96	0.93	146
C	0.97	0.98	0.98	142
D	0.94	0.97	0.96	152
E	0.96	0.97	0.97	145
F	0.98	0.97	0.98	150
G	0.95	0.98	0.96	149
H	0.95	0.92	0.94	141
I	0.99	0.94	0.97	105
J	0.96	0.99	0.97	143
K	0.95	0.99	0.97	144
L	1.00	0.96	0.98	134
M	0.99	0.97	0.98	146
N	0.99	0.97	0.98	138
O	0.98	0.94	0.96	144
P	0.97	0.98	0.98	156
Q	0.97	0.97	0.97	152
R	0.95	0.95	0.95	147
S	0.98	0.99	0.98	146
T	1.00	0.98	0.99	150
U	0.98	0.99	0.98	154
V	0.99	0.98	0.98	141

W	1.00	0.99	0.99	145
X	0.99	0.97	0.98	136
Y	0.99	0.99	0.99	149
Z	1.00	0.97	0.98	128

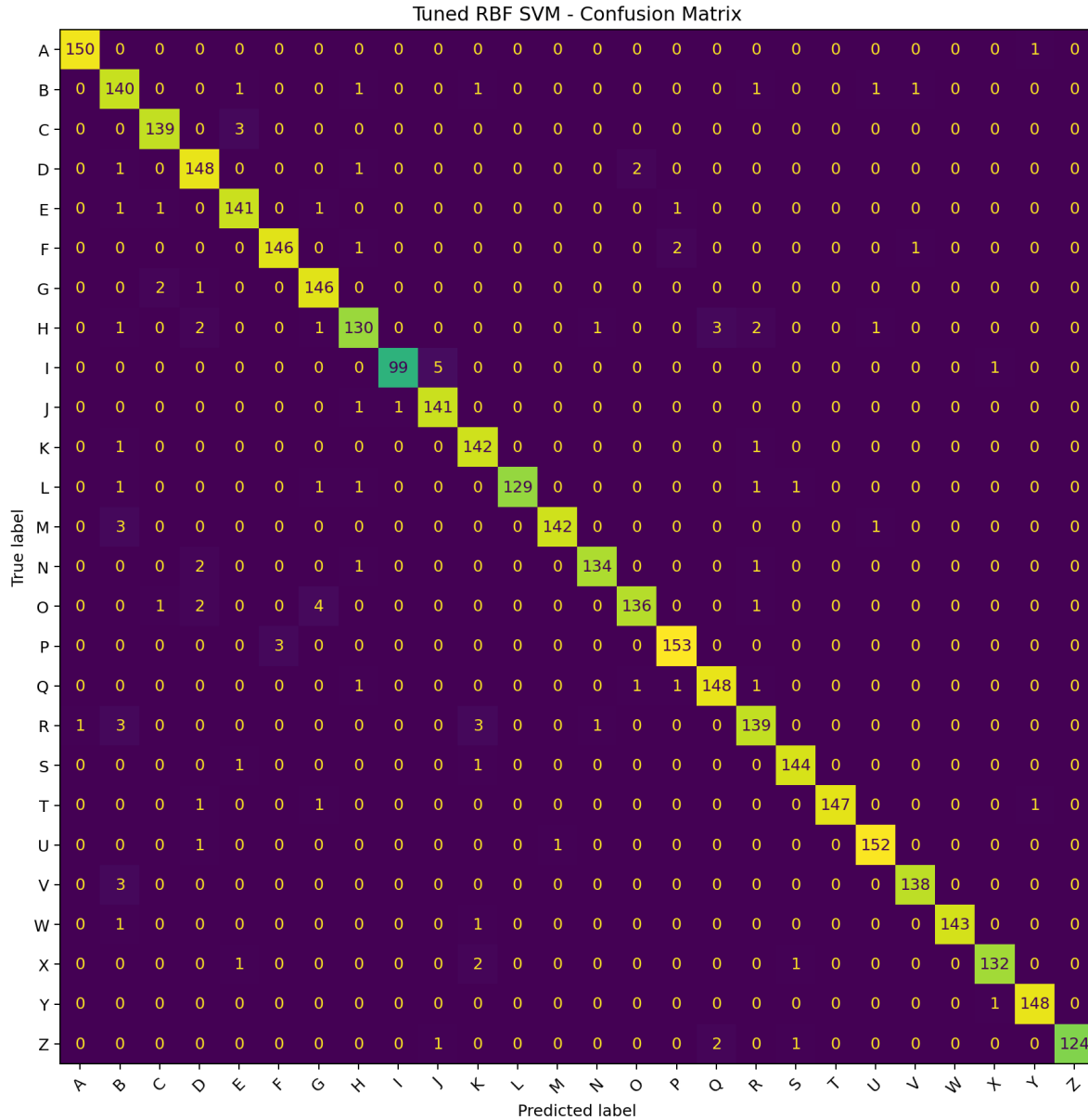


Figure 2. Confusion matrix for the tuned RBF SVM on the test set.

The diagonal cells represent correct predictions. Most values are concentrated on the diagonal, which is consistent with the 97.24% test accuracy. The small off-diagonal values show the remaining mistakes between visually or numerically similar letter patterns.

11. Discussion

The experiments show that the choice of SVM kernel has a clear effect on performance. The RBF kernel performed much better than the Linear and Polynomial configurations tested with $C=1$. This suggests that the classes are not separated well by simple linear decision boundaries. The PCA experiment reduced the feature

space from 16 dimensions to 12 components, but its accuracy was lower than the standard scaled feature set. Therefore, dimensionality reduction did not improve this SVM configuration.

The hyperparameter tuning stage produced the largest improvement. Increasing C to 10 and using $\gamma=0.1$ allowed the RBF model to create a more suitable nonlinear boundary for the data. The final test accuracy of 97.24% is also close to the 96.56% cross-validation score, suggesting that the selected configuration generalizes well to the held-out test data.

12. Limitations

- Only a small number of C and γ values were tested.
- Only three common kernels were compared in the initial experiment.
- SVC can become slow and memory intensive on much larger datasets.
- The model depends on the numerical features produced by the preprocessing pipeline and does not directly process raw letter images.
- SVM decision boundaries are less interpretable than simple rule-based or tree models.

13. Possible Improvements

- Test a wider but still controlled range of C and γ values.
- Use `RandomizedSearchCV` if a much larger hyperparameter search is needed.
- Compare additional feature-selection methods with the full scaled features.
- Test other PCA component counts rather than only the existing 12-component version.
- Compare the final SVM result with the optimum models implemented by the other group members.

14. Conclusion

This individual project part successfully implemented and evaluated a Support Vector Machine for multiclass English letter recognition. Linear, RBF, and Polynomial kernels were compared, and the RBF kernel gave the strongest initial result. RBF on PCA data was also tested but did not outperform the full scaled feature set. A small `GridSearchCV` with 3-fold cross-validation selected $C=10$ and $\gamma=0.1$ for the RBF SVM. The final tuned model achieved 97.24% test accuracy with strong precision, recall, and F1-scores across the 26 classes. Therefore, the tuned RBF SVM is selected as the final SVM configuration for this individual contribution.